



Operational research

Task A: Packing

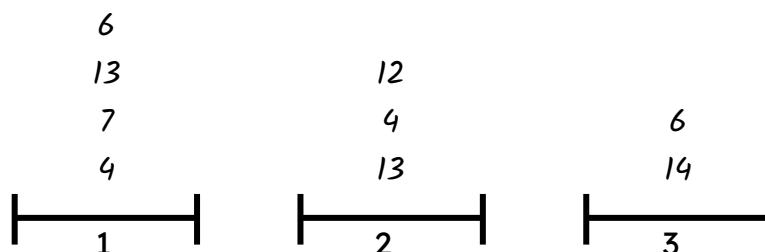
1) A ferry transports vehicles across a river. The ferry has three lanes that vehicles can use. Each lane is 30 m long.

One morning these vehicles arrive in this order:

Arrival order	Vehicle	Length (m)
1	car	4
2	car + trailer	7
3	lorry	13
4	van	6
5	lorry	13

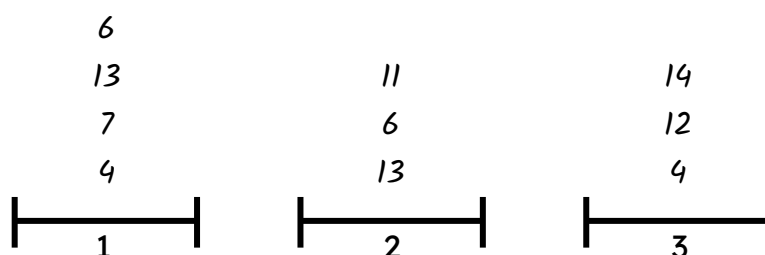
Arrival order	Vehicle	Length (m)
6	car	4
7	lorry	12
8	lorry	14
9	van	6
10	lorry	11

a) By loading each vehicle, as it arrives, into the first available lane, will all the vehicles be able to cross? Justify your answer.



The final lorry cannot make the crossing.

b) By considering all the vehicles (regardless of when they arrived), can all vehicles be loaded onto the ferry? Justify your answer.



A possible solution

c) Why is your answer to part b not necessarily practical?

E.g. Vehicles may not arrive at the same time.

It might not be possible for vehicles to pass each other in a queue.

2) Jacob is backing up his computer. He is copying games onto 32GB USB sticks. The size of each game is listed in the table.

Game	A	B	C	D	E	F	G	H
Size (GB)	8	13	17	21	22	24	25	25

a) Jacob thinks that it is possible to store all the games on five USB sticks. Explain why Jacob might think this and why he is mistaken.

$$8 + 13 + 17 + 21 + 22 + 24 + 25 + 25 = 155$$

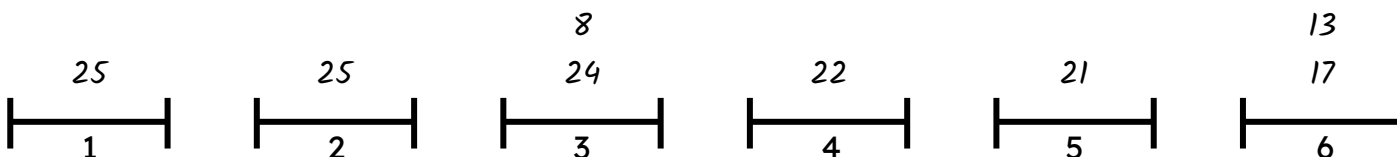
$$155 \div 32 = 4.84375 \quad \text{Therefore five USB sticks.}$$

However there are six games whose size is over half of what one USB stick can hold. These games can't be stored together meaning at least six USB sticks will be needed.

b) State the smallest number of USB sticks that will be needed and give a possible solution as to which games will be stored on each USB stick.

Six USB sticks will be needed.

E.g.



Task B: Planning a project

1) Izzy and Laura are working on a school project. They have broken the project down into a series of tasks. They create a table listing the tasks, how long it will take to complete each task and which tasks have to be completed before a task can be started.

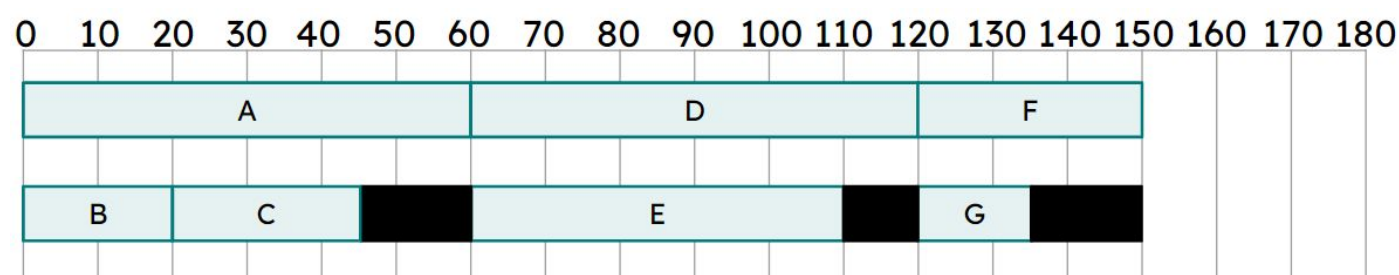
They have 2.5 hours to work on their project.

Create a chart showing who will work on which tasks and when.

Can Izzy and Laura complete their project in time?

(A template chart is provided.)

Activity	Duration	Activities which must be completed first
A (research)	60 mins	None
B (gather supplies)	20 mins	None
C (create templates)	25 mins	B
D (create artwork)	60 mins	A, C
E (writing)	50 mins	A, C
F (checking)	30 mins	D, E
G (compiling)	15 mins	D, E



This example shows Izzy and Laura completing the project on time.

